

State of the Art and Development Trends for Inspection Robots Applied in Substations

Li Li*

Department of Science and
Technology Innovation, State Grid
Intelligence Technology Co., Ltd.,
Jinan, China
lili_sdjn@163.com

Ke Xu

Development Planning Department,
State Grid Heze Power Supply
Company, Heze, China
hz_xuke@163.com

Fei Ye

Department of Science and
Technology Innovation, State Grid
Intelligence Technology Co., Ltd.,
Jinan, China
149469936@qq.com

Zhichao Liu

Department of Science and
Technology Innovation, State Grid
Intelligence Technology Co., Ltd.,
Jinan, China
zhichao_liu0219@163.com

Yanhong Liu

Department of Science and
Technology Innovation, State Grid
Intelligence Technology Co., Ltd.,
Jinan, China
liuyanhongsd@sina.cn

Hao Hu

International Standards Department,
Institute of Technology and
International Standard Co., Ltd.,
Beijing, China
hu.hao@itis-bj.com

ABSTRACT

This document introduces the state of the art of substation inspection robot which is an autonomous robotic system for inspecting equipment in substations. An overview of the existing robotics is presented, including robotics from China, the USA, New Zealand, Japan and France. And some common technology characteristics are analyzed to identify the needs for the utilities and international standards development. At the end, development trends are presented based on the market needs and technology development.

CCS CONCEPTS

• **Hardware** → Communication hardware, interfaces and storage; Sensor devices and platforms.

KEYWORDS

substations, inspection, robotics, standardization

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1 INTRODUCTION

Electric utilities are always looking for new operation and maintenance ways to achieve higher levels of safety, reliability, and efficiency. Workforce shortage is also a challenge in many countries. New technical approaches are expected to address these

issues. Traditional life-cycle management of the electric power system requires significant manpower and have issues with efficiency, consistency, quality, and especially safety in high voltage environments. Application of robotics in the electric power system is one of the promising ways to achieve these goals. Based on the rapid development of sensors and artificial intelligence, many types of robots have been developed in different countries, which could assist or replace people for different work in construction, patrol, inspection, operation, and maintenance of power facilities, either in power generation, transmission, or distribution. Currently, some types of robot systems have already been in full operation in substations for example in China [1] [2].

For application of robotics in substations, the studies carried out by the CIGRE B3.47 working group show that, thousands of unmanned ground vehicle (UGV)-based patrol robots have been in full operation in 110 kV to 1000 kV substations in China. Carrying HD cameras, infrared imagers, optional ultraviolet imagers and ultrasonic sensors, these robots can perform equipment status recognition, thermal defect detection, audible sound/noise measurement, and detection of partial discharge [3] [4]. In addition, robots packed with high-resolution cameras have been used for fault response and scheduled patrol in New Zealand, while for partial discharge detection and infrared temperature measurements in the USA. And, new inspection robots are in the research or trial operation stage in Canada and Japan. In addition, transformer internal inspection robots have been developed and used to do inspection in several countries [5], without the need to drain the oil. For equipment maintenance, live washing robots and live brushing robots have been put into trial operation in China. Other types of robots, such as operation robots and fire-fighting robots [6] [7] [8], are still under development and testing in China, Japan, and the USA.

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2 ROBOT SYSTEMS IN DIFFERENT COUNTRIES

2.1 Robots in China

In China the research on inspection robots for substations began from 2006, and in 2009 the SmartGuard robot was successfully applied in several substations to demonstrate its performance. Four generations of robots were developed with the intelligent visual inspection system and automatic control system. As of 2022 the State Grid Corporation of China has implemented more than 3000 robots in the substations rated from 110 kV to 1000 kV. Robots equipped with visible light imaging and infrared thermometers can autonomously patrol substations [9] [10] [11] [12] [13]. The robots are built with a charging room for autonomous recharging and self-management of inspection tasks. Designed with functions of autonomous meter reading and equipment inspection, these robots are capable of autonomously reading various meters, identifying switch statuses, equipment appearances, and conducting infrared temperature measurements. This replaces traditional manual inspections and enhances inspection quality and efficiency. Utilizing a 3D laser navigation system, robots can navigate within substations without being affected by minor environmental changes, and could also plan inspection paths on three dimensional maps.

Adaptability to Outdoor Environments: Robots possess a certain degree of adaptability to outdoor environments, including light rain, snowy conditions, slopes, and shallow water. Robots with four-wheel drive platforms can easily navigate through such conditions. **Challenges and Improvements:** Despite improvements in reliability, challenges persist, such as communication interruptions caused by electromagnetic interference, aging of components in outdoor environments, and the need for manual intervention when robots fail to return to the station after a malfunction. Overall, significant improvements in system reliability have been achieved through technological advancements by State Grid Intelligence Technology Co., Ltd, (shown in Fig. 1) including:

- **Enhancing Stability through Integrated Positioning and Navigation:** The 32-line LiDAR improves perception capabilities, reducing road blind spots to less than 0.5 meters. New features include global re-localization, automatic map updates, and autonomous decision-making and planning functions.
- **New Control Algorithms for Increased Intelligence:** Robots are designed with multiple self-handling logic for abnormal conditions to enhance intelligence. Pan & tilt servo algorithms are optimized to ensure precise recognition.
- **Optimized Power Supply Design for Increased Reliability:** Redundant modules are utilized in power supply components having a lifespan of over 5 years. A 100% success rate can be achieved for recharging after a failed attempt.
- **Optimized Body Structure for Enhanced Environmental Adaptability:** The entire machine is protected with an IP55 rating, making it resistant to heavy rain environments. The four-wheel drive chassis enables omnidirectional movement, with obstacle-crossing capability increased to 8 cm. Modular design enhances ease of assembly and disassembly.



Figure 1: Reliability improvement of the new inspection robots from SGAI



Figure 2: Quadruped robot for substation inspection

And with the improvement in the sensing technology and mobile platform, a new type of robot with four-legs has been tested in some substations.

The quadruped robot (shown in Fig. 2) for substation inspection carries a visual camera and an infrared thermal imager which can automatically detect instrument readings, switch status, clamp position, indicator light on/off state, and equipment temperature, thereby detecting defects before failures and reducing manual work.

The robot moves with a quadruped platform [14] [15], which has better terrain adaptability compared to wheeled robots. It adopts 3D LiDAR navigation, and can achieve forward, backward, turning, and lateral movement. It uses trot gait, with a maximum speed up to 1.2 m/s. It can not only walk on ordinary roads, but also on gravel roads. When wheeled robots encounter obstacles and need to stop on debris and steps, the quadruped robot can now directly step over without stopping. Better terrain adaptability expands the range of movement during inspections, and better observation positions can be selected when observing instrument equipment. In addition, for some roads that require detours, it can be directly traversed to shorten the detour distance and improve inspection efficiency.

With the continuous maturity of unmanned aircraft system (UAS) technology, low-cost and highly reliable multi-rotor small drones have become a common configuration for power line inspections. Considering their lower cost and wider range of information gathering methods, drones are also being used to inspect substations [16]



Figure 3: Emergency response robot (New Zealand)

[17]. However, due to constraints such as equipment positioning and safety distances, some technical challenges need to be resolved for the application of the UAS-based inspection systems working inside substations.

With the growing demand for unmanned and minimally-manned operation and maintenance models, the control mode of UAS inspection system in power equipment inspections is gradually evolving from conventional manual control to automated inspection.

When traditional UAS-based inspection systems are used for power line inspections, they require manual deployment, charge or recharges at the work site. This mode needs highly qualified operators and limits the frequency of inspections, making daily operations impractical. The intelligent drone dock, equipped with environmental and equipment status sensing, remote automatic operation system, system self-recovery from anomalies, and precise hardware fault location functions, provides a platform for autonomous drone flight, automated storage, and automated charging. This enables unmanned, automated drone inspections, meeting the requirements for remote autonomous inspections and can be utilized for unmanned autonomous patrol operations in scenarios such as substations.

2.1.1 Robots in the USA. The substation inspection robot developed in the USA carries the infrared thermal imager and a partial discharge (PD) detector, designed to detect equipment temperature and plot PD intensity distribution diagram for the whole substation defect diagnosis [18].

2.2 Robots in New Zealand

In New Zealand, Massey University has conducted pilot research on robotics for substations for years. Recently, Wrybill Robotics (shown in Fig. 3) has continued to develop the remote inspection robot which could conduct routine inspections of substation infrastructure. One robot has been operational with Transpower NZ for two years [19].

2.3 Robots in Japan

In Japan, an inspection robot (shown in Fig. 4) was developed to reduce the frequency of patrolling unmanned substations. With a high-definition camera and a laser scanner, the robot could check for anomalies, specifically focusing on oil leakage under transformers.

2.4 Robots in France

In France, two types of inspection robots (shown in Fig. 5) have been used in substations. One type, designed with 5G connection, could



Figure 4: A patrol robot from Japan



Figure 5: Two types of inspection robots (France)

conduct video surveillance of the primary medium voltage (MV) substations with real-time transmission of images [20]. The other type is developed with in-depth state diagnosis to better analyse what actions to take and/or to prepare for installation, maintenance, and repair work.

2.5 Common Composition and Key Technology

A typical patrol robot working in substations is usually constructed with four parts, i.e. the mobile platform, inspection system, control and human-machine interface (HMI) system, communication system and other auxiliary devices. There are several different platform structures, such as rail-mounted, wheeled, tracked platforms, and UAS. The control system enables the robot to travel by remote controlling or along a pre-programmed path. Some intelligent robots could do the location and navigation automatically with sensors like laser or vision-based technology. The inspection system consists of an infrared thermal imager, a visible light camera, a pan & tilt and an image processing module. The online infrared thermal imager is used to monitor hot spots in substations. The visible light camera is used to take the images of the electrical equipment in substations. Based on the image processing and pattern recognition technologies, the robot could acquire the conditions of the equipment intelligently, which will ease the burden of workers.

Based on almost 20 years of development, as shown in the table 1, shows the technological evolution for the key techniques of the robotics applied in substations.

Table 1: Key Technological Improvement of Robots Applied in Substations

Key Technology	Improvement		
	Phase I	Phase II	Phase III
Mobile platform	Rail-based	Wheel type	UAS-based
Autonomous control	GPS/Magnetic trajectory	2D/3D laser	2D/3D laser+vision
Communication technology	Wire communication	WiFi	5G
Charging technology	Wired charging	Autonomous insertion and with drawal	Wireless charging
Inspection technology	Image processing based recognition, infrared measurement	Deep learning based fault identification	Ultraviolet recognition, voiceprint recognition
Interaction mode	Manual control	human-computer interface	VR

3 DEVELOPMENT TRENDS

At the end, we give an expectation for the future of the robotics applied in the substations. With reduced cost and improved reliability, in the future, robotics will become a viable, effective, and safe solution to replace people in the hard and dangerous work, and this will be an important trend in the electric power sector.

3.1 Better Performance for Robots Application

As a developing technology, robotics for the electric power system make a sustainable development, with the new technology and new application needs. The robot systems have been used in the substations, with high failure rate, including the problems from the software, the location system, the inspection system and some others. Throughout the long time implementation, the performance of the inspection robots in substations was collected and analyzed. Technical improvement for better environmental adaptability and stability of the robots for the has been carried out in the subsidiaries of China. Stability improvement focuses on the system protection, navigation and location system, high inspection rate, automatic charging system, and more robust software system. Hardware upgrade helps the robots with better sensory ability, like the infrared camera.

3.2 Coordinated Inspection with UGV-based Robots and Drones

There are differences in the inspection range and mode between ground-based mobile inspection robots and drones. Based on the daily operation and maintenance needs of substations, automatic inspection using various types of robots can be carried out in coordination with manual inspections.

UGV-based inspection robots walks on the ground to complete such inspection tasks as meter reading, temperature measurement, equipment appearance inspection, and voiceprint recognition. Drones are used to collect visual images and infrared thermal images over equipment and equipment bays. Thanks to the larger visual field, drones can complete inspection of equipment conditions, busbar connection status, and foreign matters on the top of transformers. The two types of robotics can supplement each other in the coordinated inspection of substations.

3.3 Becoming Part of the Smart Grid

With the continuous advancement of intelligent control technology in power grid systems, techniques such as equipment status identification and defect detection based on multi-source data are being utilized across various levels of information platforms within the grid. Substation inspection robots and drones have become remote intelligent collection terminals, aiding in the accumulation of more equipment status information and supporting the construction of intelligent management and control systems for power grid operation and maintenance.

For example, currently robots can collect visual images, infrared thermal images, and audio signals of substation equipment. Some institutes have already started to research a method to identify faults in equipment conditions based on multi-modal data analysis [21] [22] [23]. As related technologies become maturer, they will be implemented in robots to realize intelligent diagnosis of equipment conditions.

3.4 Standardization Makes Technology Easier to Promote

Through the analysis of the current state of robot technology in various countries, it is evident that despite the influence of different operation and maintenance management models, the technical requirements for substation inspection robots do not vary significantly across countries. These requirements are generally based on the collection and analysis of on-site information via mobile platforms to replace and assist manual inspections with autonomous patrols.

Currently, the International Electrotechnical Commission (IEC) has established the Technical Committee 129 [24] to develop international standards on robotics used in generation, transmission, and distribution systems. For substation inspection robots, considering the existing technologies and products in various countries, the emphasis can be on the basic module composition required for robots to perform autonomous inspections within substations. This includes focusing on the functions and safety requirements of related mobile platforms, data collection devices, and control systems. At the present stage, particular attention should be paid to the accuracy of visible light-detected equipment status and meter reading identification, as well as infrared temperature detection.

Based on the analyse, environmental adaptability, inspected items, inspection time, control model, communication security, the accuracy of intelligent inspection and analysis algorithm are all the common concerns for robots applied in substations.

4 CONCLUSION

Robotic technology, especially substation inspection robots, serves as an effective means to assist or replace manual labor in power system operations and maintenance. As the technology matures, it will become an essential tool for power grid maintenance. Currently, the development of robotic technology faces challenges such as improving reliability, enhancing the service capabilities of robots, and achieving a unified application model. This requires the joint efforts of standardization organizations, users, and manufacturers. With the continuous advancement of robotic technology and the maturation of related application models, robots will welcome greater market demand in the electric power sector.

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